DISORDERS OF COLLAGEN

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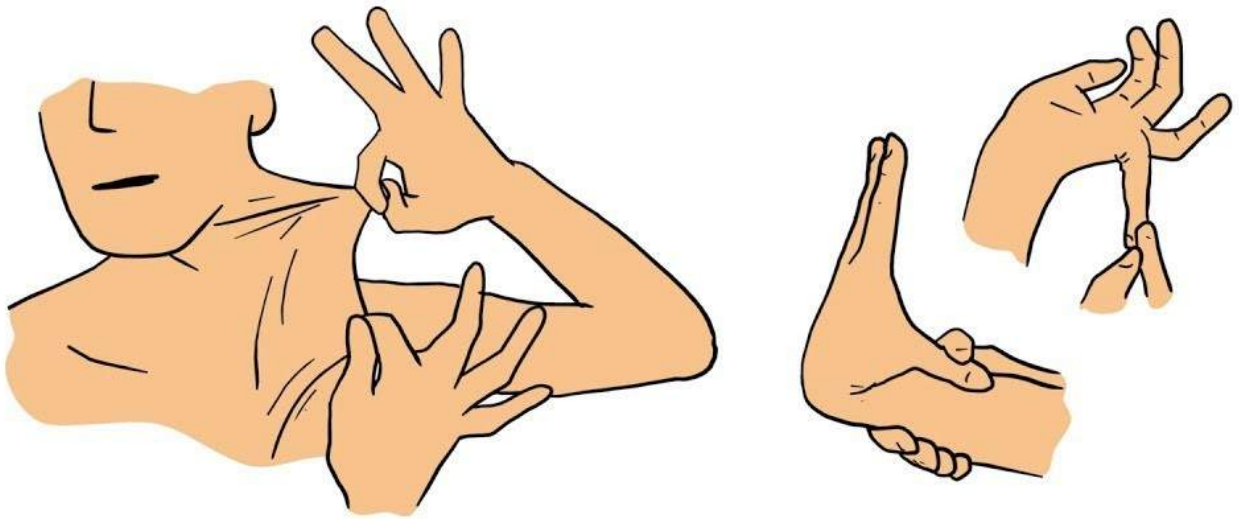
Collagenopathies

Collagen disorders may be

- Genetic
- Autoimmune
- Miscellaneous e.g. nutritional, drug-induced.

- All these affect biosynthesis assembly, post-translational modification, secretion, or other process involved in normal collagen production.


Syndrome



Ehlers-Danlos syndrome

- Ehlers-Danlos syndrome (EDS) is a heterogeneous group of connective tissue disorders that result from inheritable defects in the metabolism of fibrillar collagen molecules.

- EDS can be caused by a deficiency of collagen-processing enzymes or from mutations in the amino acid sequences of collagen types I, III, or V.

- The classic form of EDS, caused by defects in type V collagen, is characterized by skin extensibility and fragility and joint hypermobility.

- The vascular form, due to defects in type III collagen, is the most serious form of EDS because it is associated with potentially lethal arterial rupture.


A.


B.


C.
R


A


B




C

Osteogenesis imperfecta (OI):

- This syndrome, known as brittle bone disease, is a genetic disorder of bone fragility characterized by bones that fracture easily, with minor or no trauma.

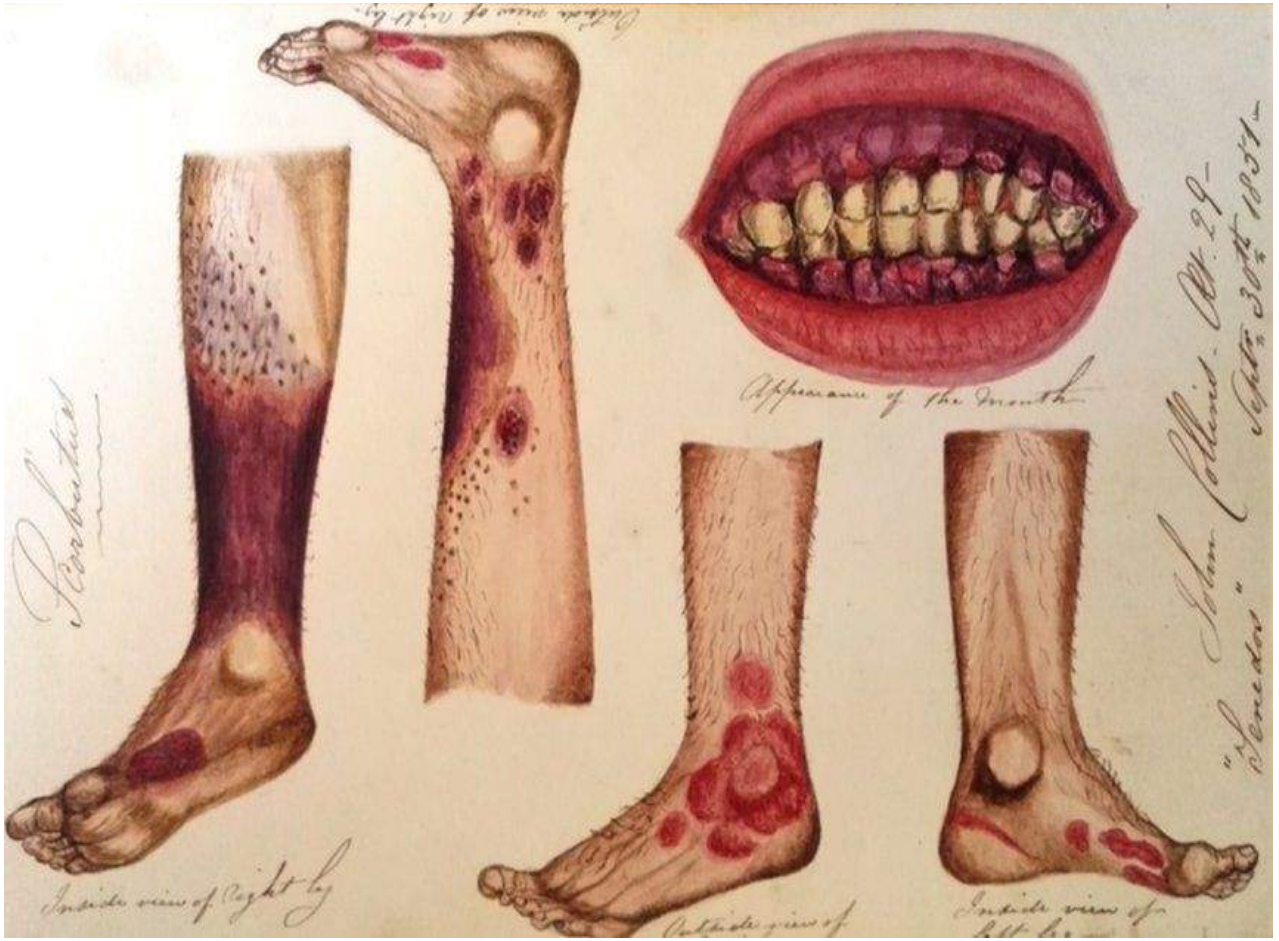
- Over 80% of cases of osteogenesis imperfecta (OI) are caused by dominant mutations to the genes that code for the α1 or α2 chains in type I collagen.

- Phenotypic severity ranges from mild to lethal.
  - **Type I:** the most common form, is characterized by mild bone fragility, hearing loss, and blue sclerae.
  - **Type II**: the most severe form, is typically lethal in the perinatal period as a result of pulmonary complications. In utero fractures are seen.
  - **Type III:** is also a severe form. It is characterized by multiple fractures at birth, short stature, spinal curvature leading to a “humped-back” (kyphotic) appearance, and blue sclerae.
  - **Dentinogenesis imperfecta**: a disorder of tooth development, may be seen in OI.

DENTINOGENSIS IMPERFECTA

- Teeth that are translucent and discolored (most often blue-grey or yellow-brown in color.
- Teeth are weaker than normal, which leads to wear, breakage, and loss of teeth.




Appearance of the mouth
Inside view of Right Leg
Inside view of

Scurvy

- The hydroxylation reactions require the reducing agent vitamin C (ascorbic acid).

- In the case of ascorbic acid deficiency, interchain formation is impaired, as is formation of a stable triple helix. Additionally, collagen fibrils cannot be cross-linked, greatly decreasing the tensile strength of the assembled fiber.

- The resulting deficiency disease is known as scurvy, a disease characterized by sore and spongy gums, loose teeth, fragile blood vessels, swollen joints, fatigue, and a microcytic anemia caused by decreased absorption of iron.

- Patients with ascorbic acid deficiency also often show bruises on the limbs as a result of subcutaneous extravasation (leakage) of blood due to capillary fragility.


Ulcer
Scar Band
Scar Band

Oral sub mucous fibrosis(OSMF)

- It is a collagen metabolic disorder with an **overall increased collagen production and decreased collagen degradation**, resulting in **increased collagen deposition** in the oral tissues and fibrosis.

- Oral Sub mucous fibrosis (OSMF) has traditionally been described as “a chronic, insidious, scarring disease of the oral cavity, often with involvement of the pharynx and the upper esophagus”.

- Although the etiopathogenesis of this disease is multifactorial, areca nut-chewing in any formulation is considered the main causative agent.

- Use of areca nut and tobacco can induce fibroblast proliferation and collagen synthesis and may penetrate the oral mucosa to cause **progressive crosslinking of collagen fibers.**

- The most common initial symptoms and signs are a
  - burning sensation
  - dry mouth
  - blanching oral mucosa, and
  - Ulceration
  - trismus

GINGIVAL ENLARGEMENT

1. **Hereditary gingival fibromatosis (HGF):**

- HGF is a rare disorder characterized by a benign, non-hemorrhagic, fibrous gingival overgrowth.

- Both dominant autosomal inheritance and recessive autosomal inheritance.



2. Drug induced enlargement:

- **Phenytoin:**
  - **An anti epileptic drug.**
  - Fibroblasts become sensitive to phenytoin, and this results in subsequent increased production of collagen.
  - The enzyme collagenase secreted by phenytoin-sensitive fibroblasts is relatively inactive to degrade collagen.
  - An imbalance in production and degradation results in the over accumulation of collagen and hence in an increase in the bulk of connective tissue.

➢ Calcium channel blockers:

After an interaction between nifedipine and gingival fibroblasts, **overproduction of collagen and extracellular ground substance occurs and** leads to an increase in the size of the gingiva.

➢ Cyclosporin:

It could react with a phenotypically distinct subpopulation of **gingival fibroblasts to enhance protein synthesis.**



Scenario 1

- A 2-month-old male baby was brought to the Orthopedic Outpatient Department with complaints of **deformities of both lower limbs following birth**.
- There was no history of trauma and similar complaints in the rest of the 2 older siblings. The baby appeared **abnormal with bilateral bowed (curved) lower limbs**. However, his **serum alkaline phosphatase** was slightly high. The x-ray of the lower limbs is shown in Figure



An x-ray of both lower limbs - anteroposterior view showing **multiple fractures** in the long bones of the lower limbs at different stages of healing, with **excessive callus formation** (arrows). Radiograph also showed **flaring and bowing of the ends of the long bones**.

What is the diagnosis?

OSTEOGENESIS IMPERFECTA (OI)

- the presence of multiple fractures of the long bones

- flaring and bowing of the ends of the long bones

What are the associated abnormalities?

- hearing loss
- blue sclera

Dentinogenesis imperfecta are the associated abnormalities, which one can encounter in patients.



Scenario 2

- A 10-year-old boy reported to our college with chief complaint of inability in chewing caused by missing teeth since past 2 years.

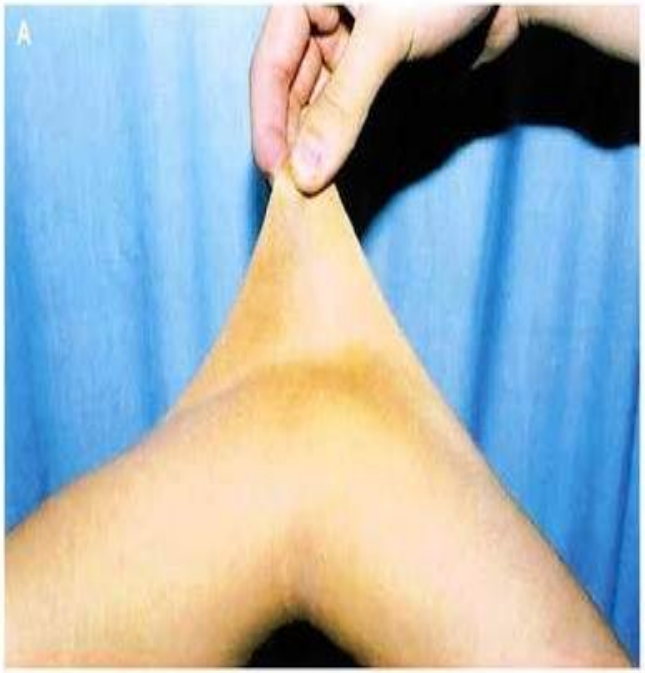
- Patient has had **joint hypermobility history since childhood.**

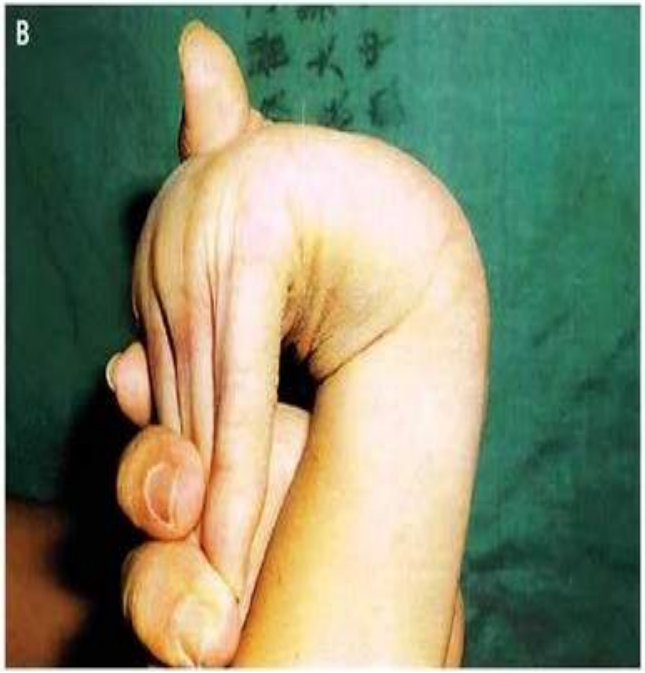
- Physical examination revealed **hyperelasticity of the skin.**

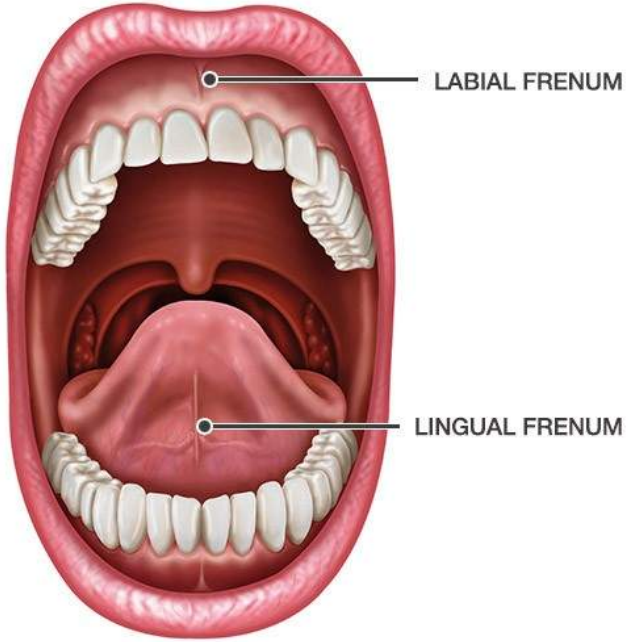
- On oral examination, **absence of the inferior labial and lingual frenula** has been reported.

What is the diagnosis?

- **EHLER DANLOS SYNDROME**
  - Joint hypermobility
  - Hyperelasticity of the skin
  - Absence of inferior labial and lingual frenula
  - Premature exfoliation of posterior deciduous teeth within five years and unerupted permanent teeth.









Absent lingual frenulum.

Scenario 3

- A 32-year-old male patient reported to the department with a **burning sensation in the mouth**.
- He did not provide any significant medical history.
- On intraoral examination, **fibrous bands were palpable** bilaterally and **tongue protrusion was significantly reduced**.
- His mouth opening was 26 mm.

- Diagnosis: Oral sub mucous fibrosis

Scenario 4

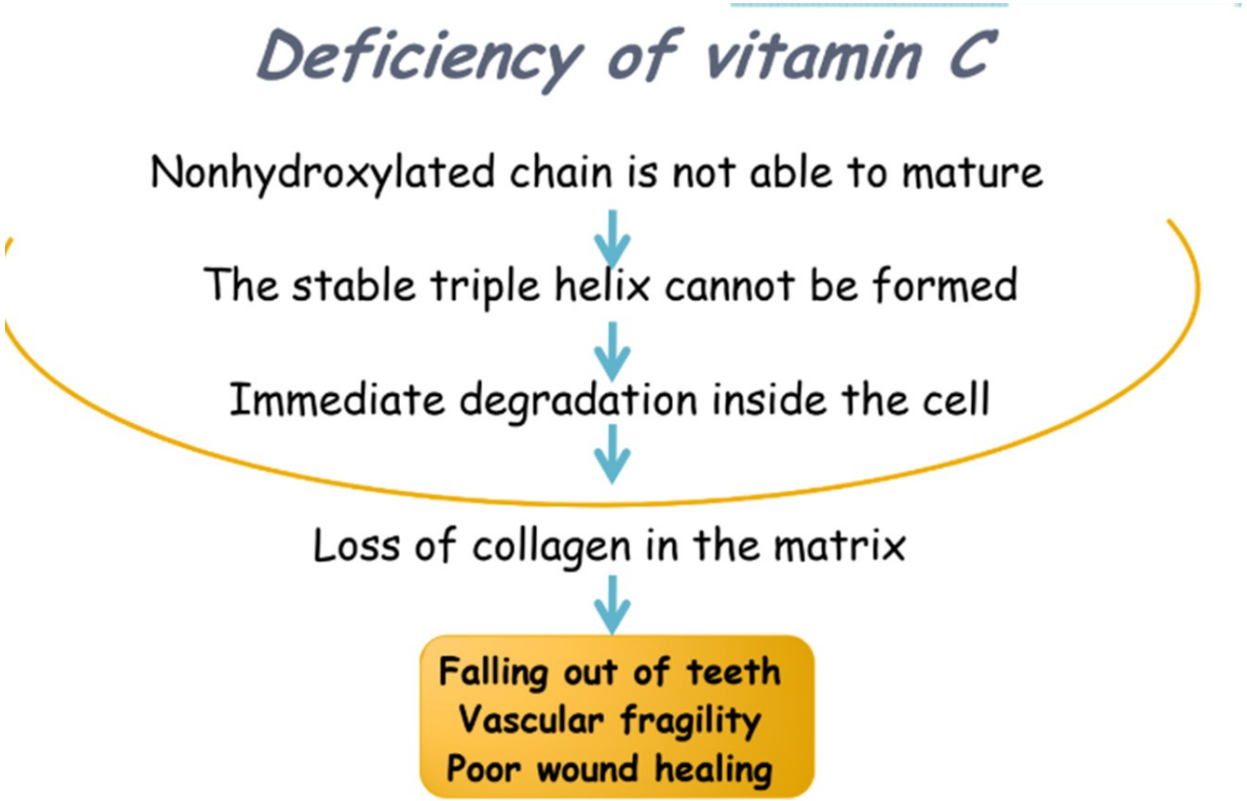
- 4 year old male pt was brought in for consultation exhibiting possible anemia, edema in his right leg, bleeding gingiva (signs of tooth loss visible), irritability, paleness, sweating, dry/cracked hair, and apparent signs of malnourishment.

- Patient is afebrile with no signs of organomegaly.

- Pt is the 10th born child of a low income household. This family has experienced infant mortality previously in the form of malnutrition and dehydration.

What is the diagnosis?

- **SCURVY**



THANK YOU